

Equivalence Class

Let $*$ be a group action of a group G on A

$$*: G \times A \rightarrow A$$

for $a, b \in A$, define a relation $\sim$ on A as

$$a \sim b \iff a = g * b, \quad \text{for some } g \in G$$

then $\sim$ is an equivalence relation on A .

Proof:

$\forall a \in A$

$$a = e * a, \quad e \in G$$

$$\Rightarrow a \sim a$$

$\therefore \sim$ is reflexive.

Let $a, b \in A$ s.t. $a \sim b$

$$\Rightarrow \exists g \in G \text{ s.t.}$$

$$a = g * b$$

$$\Rightarrow g^{-1} * a = g^{-1} * (g * b)$$

$$\Rightarrow g^{-1} * a = b$$

$$\Rightarrow b \sim a$$

$\therefore \sim$ is symmetric.

Let $a, b, c \in A$ s.t. $a \sim b$ and $b \sim c$

$$\Rightarrow \exists g_1, g_2 \in G \text{ s.t.}$$

$$a = g_1 * b \quad \text{and} \quad b = g_2 * c$$

$$a = g_1 * (g_2 * c)$$

$$a = (g_1 g_2) * c \quad (g_1 g_2 \in G)$$

$$\Rightarrow a \sim c$$

$\therefore \sim$ is transitive and equivalence relation.

So, it will partition the set A into its equivalence class.

Let $a \in A$

$$\begin{aligned}\text{Cl}(a) &= \{b \in A \mid a \sim b\} \\ &= \{b \in A \mid b = g * a, \forall g \in G\} \\ &= \{g * a \mid g \in G\}\end{aligned}$$

$$\text{Cl}(a) = G \cdot a \quad (\text{orbit of } a \text{ under } *)$$

$$\Rightarrow A = \bigcup_{a \in A} \text{Cl}(a)$$

$$\Rightarrow A = \bigcup_{a \in A} G \cdot a$$

Conjugacy class

If G acts on itself by conjugation, then

$$G_a = \text{Normalizer of } a = N(a)$$

$$\begin{aligned}G \cdot a &= \text{Cl}(a) = \{g * a \mid g \in G\} \\ &= \{gag^{-1} \mid g \in G\} \quad (\text{conjugacy class of } a)\end{aligned}$$

H.W

Let G be a group and $a, b \in G$. Then a and b are conjugate if $\exists c \in G$ such that $b = cac^{-1}$. Prove that $\sim$ is an equivalence relation on G where $a \sim b$ when a is conjugate to b .

Soln:

Let $a \in G$, then

$$a = eae^{-1} \quad (\text{as } e \in G)$$

$\Rightarrow a \sim a$ and $\sim$ is reflexive.

Let $a, b \in G$ such that $a \sim b$, then $\exists c \in G$ s.t.

$$\begin{aligned} a &= cbc^{-1} \\ c^{-1}a &= c^{-1}cbc^{-1} \\ c^{-1}ac &= ebc^{-1}c \\ c^{-1}ac &= be \\ c^{-1}a(c^{-1})^{-1} &= b \end{aligned}$$

$\Rightarrow b \sim a$ and $\sim$ is symmetric.

Let $a, b, c \in G$ s.t. $a \sim b$ and $b \sim c$
then $\exists d_1, d_2 \in G$ s.t.

$$\begin{aligned} a &= d_1bd_1^{-1} \\ b &= d_2cd_2^{-1} \\ a &= d_1d_2cd_2^{-1}d_1^{-1} \quad \{d_2^{-1}d_1^{-1} = (d_1d_2)^{-1}\} \\ a &= (d_1d_2)c(d_1d_2)^{-1} \quad \{\text{as } d_1d_2 \in G\} \end{aligned}$$

$\Rightarrow a \sim c$ and $\sim$ is transitive and hence equivalence.

Note: By orbit-stabilizer theorem $\exists$ a bijection between the set $\text{Cl}(a)$ and $G/N(a)$.

$$G = \bigcup_{a \in G} \text{Cl}(a)$$

1. If G is a finite group then

$$\begin{aligned} |\text{Cl}(a)| &= \left| \frac{G}{N(a)} \right| \\ |G| &= \sum_{a \in G} |\text{Cl}(a)| \end{aligned}$$

To Prove: $\text{Cl}(a) = \{a\} \iff a \in Z(G)$.

Proof:

Let $\text{Cl}(a) = \{a\}$.

If $g \in G$ then

$$\begin{aligned} gag^{-1} &\in \text{Cl}(a) = \{a\} \\ \Rightarrow gag^{-1} &= a \\ \Rightarrow ga &= ag \end{aligned}$$

As g is arbitrary so $a \in Z(G)$.

Conversely, let $a \in Z(G)$

$$\begin{aligned} \text{Cl}(a) &= \{gag^{-1} \mid g \in G\} \\ &= \{agg^{-1} \mid g \in G\} \\ \text{Cl}(a) &= \{a\} \end{aligned}$$

To Prove: $a \in Z(G) \iff N(a) = G$

Proof:

Let $a \in Z(G)$

$$\begin{aligned} \Rightarrow ag &= ga \quad \forall g \in G \\ \Rightarrow g &\in N(a) \quad \forall g \in G \\ \Rightarrow N(a) &= G \end{aligned}$$

Conversely, let $N(a) = G$

then $\forall g \in G, g \in N(a)$

$$\begin{aligned} \Rightarrow ag &= ga \quad \forall g \in G \\ \Rightarrow a &\in Z(G) \end{aligned}$$

Class Equation:

$$|G| = |Z(G)| + \sum_{a \notin Z(G)} |G/N(a)|$$